

Self Service Coffee Kiosk Architecture Justification

Architecture Selection

We chose Layered Architecture for the Self Service Coffee Kiosk System.

Reason 1: The kiosk has a compact, predictable workflow. Separating presentation, business logic, and infrastructure keeps touch-screen changes independent from ordering rules, payment integration, menu storage, and printer code.

Reason 2: Each hardware or external dependency is replaceable behind an interface. A different card provider, database, or printer driver can be introduced without rewriting the Order Manager or Touchscreen UI.

Security advantage: Only the Payment Service handles payment authorization. The Order Manager sends an amount and order reference, receives only the result and transaction reference, and never stores raw card data, reducing the sensitive-data boundary.

Performance benefit: In-process layer calls and a local menu repository avoid unnecessary service-to-service network hops. Menu and pricing reads remain fast, while only payment authorization uses an external connection.

Functional needs	Select 3 coffee types and 2 sizes; calculate price; accept credit-card payment; print an itemized receipt.
Technical needs	Responsive touch input, reliable printer communication, persistent menu and pricing data, and clear failure handling.
Main challenges	Usability during busy periods, payment-data protection, fast response, and graceful handling of payment or printer faults.
Components	Touchscreen UI, Order Manager, Payment Service, Menu and Pricing Repository, and Receipt Printer Adapter.

Architectural Style Analysis

Style	Scenario fit	Main advantage	Main limitation
Layered	High	Clear separation of UI, ordering, and device/data access	Some cross-layer call overhead
Microservices	Low	Independent deployment and scaling	Network, deployment, and consistency complexity is excessive for one kiosk

Client Server	Medium	Centralized data and control	Server dependency can stop ordering and adds network latency
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Diagram Traceability

The accompanying UML component diagram shows five components and four typed assembly connectors: **IOrdering**, **IMenuCatalog**, **IPaymentProcessing**, and **IReceiptPrinting**.